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MID SEMESTER EXAMINATION- SEPTEMBER 2025

Name of the Course: B.Tech

Name of the Paper : Advanced Structural Analysis

Time: 1½ Hours

Semester : V semester

Paper Code: TCE 501

Maximum Marks: 50

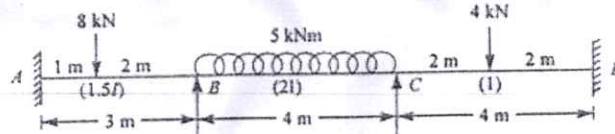
- Note: i) All questions are compulsory
ii) Each question carries 10 Marks

Q.1

(10 Marks)

- a) Determine the support moments A, B, C and D by using moment Distribution methods

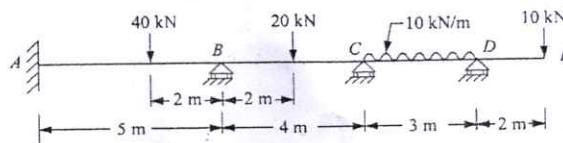
(CO 3)



OR

- b) Determine the support moments A, B, C and D by using moment Distribution methods

(CO 3)

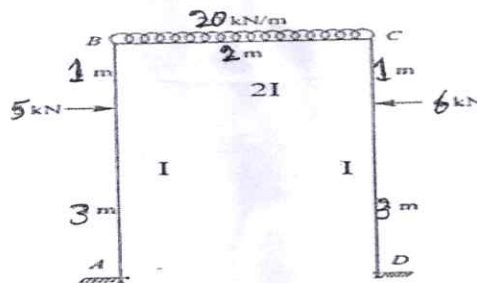


Q.2

(10 Marks)

- a) Analyze the following portal frame by Moment Distribution method. Support A and D are fixed.

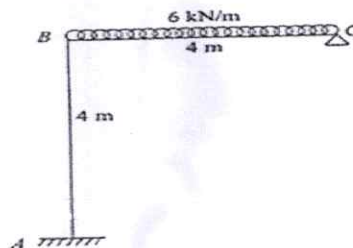
(CO 3)



OR

- b) Analyze the following portal frame by Moment Distribution method

(CO 3)

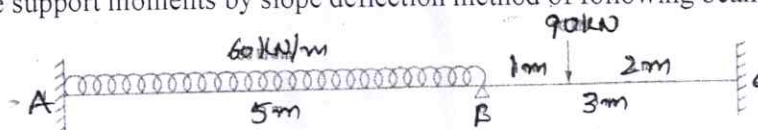


Q.3

(10 Marks)

- a) Determine the support moments by slope deflection method of following beam.

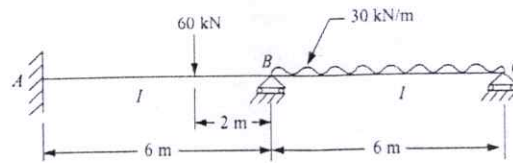
(CO 3)



OR

b) Determine the support moments by slope deflection method of following beam.

(CO 3)

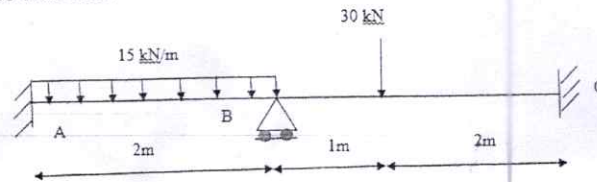


Q.4

(10 Marks)

a) Analyse the following beam by stiffness method.

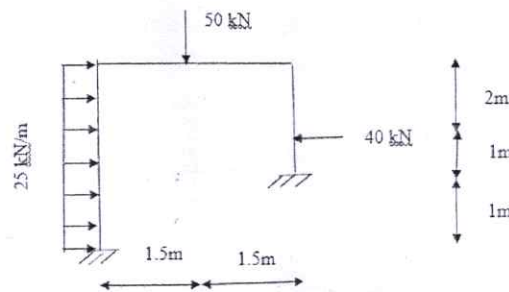
(CO4)



OR

b) Analyse the rigid jointed plane frame supported and loaded as shown in Fig by stiffness matrix method.

(CO 4)

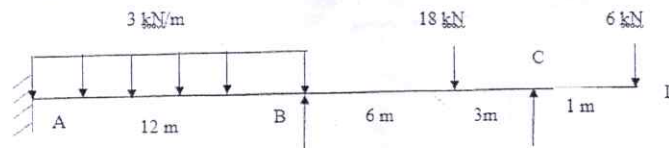


Q.5

(10 Marks)

a) Analyse the following beam by stiffness matrix method.

(CO 4)



OR

b) Analyse the following portal frame by slope deflection method.

(CO 3)

